

Sales Performance Analysis

From business problem to data-driven recommendation

Tools	Microsoft Excel SQL Power BI
Dataset	1,577 cleaned transaction rows 1,145 orders 639 customers
Period	2025-09-01 to 2026-08-29
Prepared	29 August 2026

This guide turns "Sales Analysis" from a practice exercise into a portfolio project that demonstrates the full Business Analyst arc: **business problem to clean data to analysis to visualisation to recommendation** . It ships with a generated 12-month e-commerce dataset that has real, findable patterns built into it.

NET REVENUE Rs 19,333,556	ORDERS 1,145	AVG ORDER VALUE Rs 16,885	GROSS MARGIN 23.0%
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Three corrections to the usual version of this project

- 1. Add a cost column.** A dataset with only Price cannot support any recommendation about margin or profitability - yet that is exactly the kind of recommendation these projects always make. Unit Cost and Discount % are what make the analysis defensible.
- 2. Never quote a number you did not compute.** Insights are derived at the end, from your own data. Every figure in this document is calculated from the supplied dataset at build time.
- 3. Present it as a case study, not a task list.** Recruiters skim for the narrative: problem, decision, evidence, recommendation. The same work reads as far more senior when it is framed that way.

The distinction that matters

A dashboard shows what happened. A Business Analyst explains why it happened and states what the business should do next. The second half is the entire job - and the half most portfolio projects leave out.

01 Scope it like a BA

Before touching data, write a one-paragraph project brief. This is the artifact that separates a Business Analysis project from a dashboard exercise, and it is the step almost everyone skips.

Field	Definition for this project
Stakeholder	Head of Sales, a mid-size Indian e-commerce retailer
Business problem	Revenue is flat quarter-on-quarter while order volume holds steady. Leadership cannot tell whether the cause is pricing, product mix, region or customer retention.
Decision to support	Where to allocate next quarter's Rs 5 00 000 promotional budget
Success metric	Lift average order value by 10% without reducing order count
In scope	12 months of transactional sales data (2025-09-01 to 2026-08-31)
Out of scope	Returns, marketing spend, inventory levels, competitor pricing
Assumptions	Prices are pre-tax; cancelled orders excluded; Region is the delivery city; Unit Cost is landed cost excluding fulfilment

Test your framing

"Revenue is flat and we do not know why" is a business problem. "I want to practise SQL" is not. Every chart you build later must trace back to the decision this brief names.

02 Design the data

Generating your own dataset beats downloading one, because you control the story and you are not the four-hundredth candidate to analyse the same file. The schema below matches the specification for this project, plus three analytical extensions.

Data dictionary

Column	Type	Purpose / notes
Order ID	Integer	Repeats across rows for multi-item orders – use DISTINCT when counting orders
Date	Date	Spread across 12 months with real seasonality
Customer	Text	639 customers across 1145 orders, which is what makes repeat-rate analysis possible
Product	Text	32 distinct SKUs
Category	Text	Electronics, Furniture, Home & Kitchen, Apparel, Accessories
Quantity	Integer	Right-skewed; big-ticket items sell 1-2 per order, small items sell in larger baskets
Price	Decimal	Unit list price in rupees, before discount
Unit Cost	Decimal	ADDED - unlocks margin analysis instead of revenue-only analysis
Discount %	Decimal	ADDED - unlocks discount-effectiveness analysis, the strongest finding in this dataset
Region	Text	8 cities; also the deliberate blank-value column
Channel	Text	ADDED - Web or Mobile App, a cheap extra slicing dimension

Five patterns deliberately built into the data

Clean, featureless data gives you nothing to write about. The generator plants findable structure and realistic mess:

Pattern	What was planted	What it lets you demonstrate
Pareto skew	A minority of SKUs carries most revenue	Concentration analysis, running totals, catalogue rationalisation
Seasonality	Festive spike in Oct-Nov, slump in Jan-Feb	Trend analysis, month-on-month growth, LAG window functions
Margin trap	The largest category by revenue has thin margins	Why revenue alone misleads; revenue vs profit thinking
Retention gap	One region acquires well but barely repeats	Cohort/repeat-rate analysis, acquisition vs retention trade-off

Pattern	What was planted	What it lets you demonstrate
Deliberate mess	Duplicates, blanks, casing variants, bad dates, price outliers	Data quality judgement and an auditable cleaning log

Reproduce or regenerate it

```
# Regenerate the whole dataset (fixed seed 42 = reproducible output)
python3 scripts/generate_sales_dataset.py

# Outputs written to data/
# raw_sales_data.csv      1,626 rows, uncleaned, all defects present
# cleaned_sales_data.csv  1,577 rows, cleaned + calculated columns
# Sales_Data_Raw.xlsx     the raw file as a real workbook for Excel work
# data_quality_log.csv    the audit trail of every cleaning rule
```

03 Clean it in Excel, and log every decision

Do the cleaning, but produce a **Data Quality Log** as a deliverable in its own right. Documenting your exclusions is what professional analysts get paid for, and it is the single easiest way to look experienced. The log below was generated from the actual cleaning run.

Issue found	Rows	Rule applied	Rationale
Exact duplicate rows	41	Removed	Identical Order ID + Product + Date = double entry from the source export
Inconsistent Region casing / whitespace	62	TRIM + PROPER applied	Prevented duplicate region groups splitting GROUP BY totals
Blank Region	28	Set to "Unknown"	Dropping these rows would understate national revenue totals
Mixed date formats (DD-MM-YYYY)	20	Parsed and normalised to YYYY-MM-DD	Text-format dates break chronological sorting and date logic
Negative Quantity	5	Excluded and flagged	Likely untagged returns; returns are out of scope for this brief
Order Date later than today	3	Excluded	Impossible values - cannot represent a completed sale
Price outliers (10x product median)	2	Corrected to product median price	Confirmed data-entry error, not a premium SKU variant
NET RESULT	49	1626 rows in -> 1577 rows out	3.0% of records removed

Excel techniques to name explicitly

- ? **Remove Duplicates** on the full column set, not a single column
- ? **TRIM, PROPER, CLEAN** to standardise text - this is what stops one city splitting into three phantom groups in a later GROUP BY
- ? **Power Query** for the transform steps, so the cleaning is repeatable rather than a one-off manual pass
- ? **XLOOKUP** for category and cost mapping, **IFERROR** to trap gaps
- ? **Conditional formatting** to surface price and quantity outliers visually before you decide what to do with them
- ? **Pivot tables** for a first sanity check on totals before you trust any SQL result

Calculated columns

```
Gross Revenue = Quantity * Price
Net Revenue   = Quantity * Price * (1 - [Discount %])
Gross Profit  = Net Revenue - (Quantity * [Unit Cost])
Margin %      = Gross Profit / Net Revenue
Order Month   = TEXT([Date], "yyyy-mm")
```

Interview line

Say this in the interview: "I removed 3.0% of records and documented every rule, so the analysis is reproducible and the exclusions are auditable." That one sentence signals more maturity than any chart you can show.

04 Analyse it in SQL as a question ledger

Structure your SQL file as **Question, Query, Answer, So what** rather than as a pile of statements. Anyone reading it then follows your reasoning, not just your syntax. Below are the five queries that actually differentiate you - everyone else writes `SUM()` ... `GROUP BY`.

Query 1 - Month-on-month growth (trend, not just totals)

```
WITH monthly AS (  
    SELECT DATE_TRUNC('month', order_date) AS month,  
           SUM(quantity * unit_price * (1 - discount_pct)) AS revenue  
    FROM sales  
    GROUP BY 1  
)  
SELECT month,  
       ROUND(revenue, 0) AS revenue,  
       ROUND(LAG(revenue) OVER (ORDER BY month), 0) AS prev_month,  
       ROUND(100.0 * (revenue - LAG(revenue) OVER (ORDER BY month))  
             / NULLIF(LAG(revenue) OVER (ORDER BY month), 0), 1) AS mom_growth_pct  
FROM monthly  
ORDER BY month;
```

Query 2 - Pareto concentration (which SKUs actually matter)

```
WITH product_rev AS (  
    SELECT product_name,  
           SUM(quantity * unit_price * (1 - discount_pct)) AS revenue  
    FROM sales  
    GROUP BY product_name  
)  
ranked AS (  
    SELECT product_name, revenue,  
           SUM(revenue) OVER (ORDER BY revenue DESC  
                              ROWS BETWEEN UNBOUNDED PRECEDING  
                              AND CURRENT ROW) AS running_rev,  
           SUM(revenue) OVER () AS total_rev,  
           ROW_NUMBER() OVER (ORDER BY revenue DESC) AS rn,  
           COUNT(*) OVER () AS n_products  
    FROM product_rev  
)  
SELECT product_name,  
       ROUND(revenue, 0) AS revenue,  
       ROUND(100.0 * running_rev / total_rev, 1) AS cumulative_revenue_pct,  
       ROUND(100.0 * rn / n_products, 1) AS cumulative_product_pct  
FROM ranked  
ORDER BY revenue DESC;
```

Query 3 - Category action matrix (recommendation in the output)

```
WITH cat AS (  
    SELECT category,  
           SUM(quantity * unit_price * (1 - discount_pct)) AS revenue,  
           SUM(quantity * unit_price * (1 - discount_pct))  
             / COUNT(DISTINCT order_id) AS aov,  
           SUM(quantity * unit_price * (1 - discount_pct)  
             - quantity * unit_cost)  
             / SUM(quantity * unit_price * (1 - discount_pct)) AS margin_pct  
    FROM sales  
    GROUP BY category  
)
```

```

FROM sales
GROUP BY category
)
SELECT category,
ROUND(revenue, 0) AS revenue,
ROUND(aov, 0) AS aov,
ROUND(100 * margin_pct, 1) AS margin_pct,
CASE
WHEN revenue >= AVG(revenue) OVER ()
AND margin_pct >= AVG(margin_pct) OVER ()
THEN 'Star - protect and scale spend'
WHEN revenue >= AVG(revenue) OVER ()
THEN 'Volume driver - cut discounting, bundle to lift margin'
WHEN margin_pct >= AVG(margin_pct) OVER ()
THEN 'Hidden gem - increase traffic and visibility'
ELSE 'Underperformer - rationalise or reprice'
END AS recommended_action
FROM cat
ORDER BY revenue DESC;

```

Query 4 - Discount effectiveness (the money question)

```

SELECT CASE WHEN discount_pct = 0 THEN '0%'
WHEN discount_pct <= 0.10 THEN '1-10%'
WHEN discount_pct <= 0.20 THEN '11-20%'
ELSE '21%+' END AS discount_band,
COUNT(DISTINCT order_id) AS orders,
ROUND(SUM(quantity * unit_price * (1 - discount_pct))
/ COUNT(DISTINCT order_id), 0) AS aov,
ROUND(100 * SUM(quantity * unit_price * (1 - discount_pct)
- quantity * unit_cost)
/ SUM(quantity * unit_price * (1 - discount_pct)), 1) AS margin_pct
FROM sales
GROUP BY 1
ORDER BY 1;

```

Query 5 - Repeat rate and revenue concentration by customer

```

WITH cust AS (
SELECT customer_id,
COUNT(DISTINCT order_id) AS orders,
SUM(quantity * unit_price * (1 - discount_pct)) AS spend
FROM sales
GROUP BY customer_id
),
deciled AS (
SELECT *, NTILE(10) OVER (ORDER BY spend DESC) AS spend_decile
FROM cust
)
SELECT COUNT(*) AS total_customers,
ROUND(100.0 * COUNT(*) FILTER (WHERE orders > 1) / COUNT(*), 1)
AS repeat_rate_pct,
ROUND(AVG(spend), 0) AS avg_customer_value,
ROUND(100.0 * SUM(spend) FILTER (WHERE spend_decile = 1)
/ SUM(spend), 1) AS top_decile_revenue_share
FROM deciled;

```

`FILTER (WHERE ...)` is PostgreSQL syntax. On MySQL or SQL Server substitute `SUM(CASE WHEN ... THEN 1 ELSE 0 END)`. Both variants are included in `sql/03_advanced_analysis.sql`.

What this demonstrates

Skills you can now legitimately claim: common table expressions, window functions (LAG, ROW_NUMBER, NTILE, running totals with an explicit frame clause), aggregate window functions, CASE-based segmentation, date truncation, and NULLIF for safe division.

05 Build the Power BI dashboard: model first

5.1 Build a real star schema, not one flat table

Table	Role	Contents
Fact_Sales	Fact	One row per order line: quantity, price, discount, keys
Dim_Date	Dimension	Generated with CALENDARAUTO(), then Mark as Date Table
Dim_Product	Dimension	Product, Category, Unit Cost
Dim_Customer	Dimension	Customer, home Region

Being able to say "I built a star schema with a marked date dimension so time intelligence works correctly" puts you ahead of most junior candidates, who load a single flat sheet and then wonder why TOTALYTD misbehaves.

5.2 Core DAX measures

```
Total Revenue =
    SUMX(Fact_Sales,
        Fact_Sales[Quantity] * Fact_Sales[Price] * (1 - Fact_Sales[Discount %]))

Total Cost      = SUMX(Fact_Sales,
    Fact_Sales[Quantity] * RELATED(Dim_Product[Unit Cost]))

Gross Profit    = [Total Revenue] - [Total Cost]
Margin %        = DIVIDE([Gross Profit], [Total Revenue])

Total Orders    = DISTINCTCOUNT(Fact_Sales[Order ID])
Units Sold      = SUM(Fact_Sales[Quantity])
AOV             = DIVIDE([Total Revenue], [Total Orders])

Revenue PM      = CALCULATE([Total Revenue], DATEADD(Dim_Date[Date], -1, MONTH))
Revenue MoM %   = DIVIDE([Total Revenue] - [Revenue PM], [Revenue PM])
Revenue YTD     = TOTALYTD([Total Revenue], Dim_Date[Date])

Repeat Customers =
    COUNTROWS(
        FILTER(VALUES(Fact_Sales[Customer]),
            CALCULATE(DISTINCTCOUNT(Fact_Sales[Order ID])) > 1))

Repeat Rate %   = DIVIDE([Repeat Customers], DISTINCTCOUNT(Fact_Sales[Customer]))
```

5.3 One page, ordered by decision-relevance

Not five pages of every chart you know how to make. One executive page that answers the brief, with an optional detail page behind it.

```
+-----+
| SALES PERFORMANCE           [ Date ] [ Category ] [ Region ] |
+-----+
| Revenue 1.93Cr      Orders 1,145   AOV 16,885      Margin 23.0% |
+-----+
| Revenue trend + MoM % | Category matrix: |
| (column + line, festive spike) | revenue vs margin scatter |
+-----+
| Top 10 products with Pareto | Region performance: |
| cumulative % line          | revenue bar + repeat rate |
+-----+
```

5.4 Design rules that make it look professional

- ? **Title every visual as a finding, not a label.** "Festive quarter drove 34% of annual revenue" beats "Revenue by Month".
- ? **One accent colour, grey for context.** Red and green reserved strictly for where action is needed.
- ? **Consistent rupee formatting** , no decimals on large numbers, thousands separators everywhere.
- ? **No 3D, no pie chart with nine slices, no rainbow palette.** These read as inexperience.

- ? **Put one sentence of written insight on the canvas.** A dashboard that states its own conclusion is worth three that do not.
- ? **Sort bars by value, not alphabetically** - the ranking is the point.

06 Findings, recommendations and sizing

This is the whole project. Everything before it is preparation. Use a strict three-part structure for each insight:

Observation (what the data says, with a number), **Interpretation** (why it is happening), **Recommendation** (a specific action with an expected rupee impact).

The four findings below are computed from the supplied dataset.

Finding 1 - Deep discounting is not buying bigger baskets

Discount band	Orders	Net revenue	AOV	Margin %
0%	240	Rs 3,667,000	Rs 15,280	36.1%
1-10%	461	Rs 6,290,348	Rs 13,645	29.4%
11-20%	418	Rs 5,759,795	Rs 13,779	18.8%
21%+	320	Rs 3,616,334	Rs 11,301	5.4%

- ? **Observation.** Orders discounted above 20% carry an average order value of Rs 11,301 against Rs 15,280 for undiscounted orders - 26% **lower**, not higher - while gross margin falls from 36.1% to 5.4%.
- ? **Interpretation.** Deep discounts are not persuading customers to buy more per order. They are subsidising purchases that were likely to happen anyway, converting margin into no incremental volume.
- ? **Recommendation.** Cap standard discounts at 15% and redirect the difference into threshold-based bundles ("free shipping over Rs 17,000"). Repricing the 496 affected orders at a 15% cap recovers **Rs 769,336 in gross profit**, a 17% improvement on total profit, at constant volume.

Finding 2 - The biggest category by revenue is the weakest by margin

Category	Net revenue	Revenue share	Margin %	AOV
Electronics	Rs 12,023,186	62.2%	14.9%	Rs 20,694
Furniture	Rs 4,352,378	22.5%	38.1%	Rs 17,203
Home & Kitchen	Rs 1,978,660	10.2%	26.4%	Rs 8,076
Apparel	Rs 685,250	3.5%	49.5%	Rs 2,891
Accessories	Rs 294,082	1.5%	46.1%	Rs 1,730

- ? **Observation.** Electronics generates 62.2% of net revenue at only 14.9% margin, while Apparel returns 49.5% margin on 3.5% of revenue.
- ? **Interpretation.** Ranking categories by revenue would put Electronics first and make it the obvious place to spend. Ranking by profit contribution tells a different story - it is a high-turnover, low-return category being made worse by the discounting in Finding 1.
- ? **Recommendation.** Stop treating revenue share as importance. Hold Electronics spend flat, protect it on price rather than promoting it, and shift incremental budget toward the higher-margin categories where each rupee of revenue keeps more profit.

Finding 3 - A third of the catalogue does nearly all the work

Rank	Product	Net revenue	Share	Cumulative
1	Laptop	Rs 4,199,284	21.7%	21.7%
2	Phone	Rs 2,696,564	13.9%	35.7%
3	Smart TV	Rs 2,536,568	13.1%	48.8%
4	Sofa Set	Rs 1,523,798	7.9%	56.7%
5	Tablet	Rs 1,107,197	5.7%	62.4%
6	Bed Frame	Rs 866,620	4.5%	66.9%
7	Wardrobe	Rs 782,772	4.0%	70.9%
8	Smartwatch	Rs 724,279	3.7%	74.7%

- ? **Observation.** 10 of 32 products (31% of the catalogue) generate 80% of net revenue. 16 SKUs each contribute under 1%.
- ? **Interpretation.** The long tail is not free. It consumes catalogue management, photography, storage and merchandising attention disproportionate to its contribution.
- ? **Recommendation.** Review the 16 sub-1% SKUs for delisting or supplier renegotiation, and reinvest the freed merchandising effort in the top 10 products that already drive four fifths of revenue.

Finding 4 - One region acquires customers but cannot keep them

Region	Customers	Repeat rate	Net revenue
Chennai	153	39.2%	Rs 5,308,066
Bengaluru	95	38.9%	Rs 2,481,696
Coinimbatore	79	40.5%	Rs 2,397,824
Madurai	84	40.5%	Rs 2,305,638
Salem	104	12.5%	Rs 2,168,000
Hyderabad	59	42.4%	Rs 1,999,276
Trichy	34	44.1%	Rs 1,282,820
Kochi	24	41.7%	Rs 1,019,430

- ? **Observation.** Salem has 104 customers - more than several higher-revenue regions - but a repeat rate of just 12.5% against a 41.0% average across the other regions.
- ? **Interpretation.** Acquisition in Salem is working; retention is not. The region is buying first orders and losing the customers immediately afterwards, which is the most expensive way to grow.
- ? **Recommendation.** Suspend incremental acquisition spend in Salem and run a post-purchase retention programme (second-order incentive, delivery-experience audit) before adding more top-of-funnel budget. Closing half the gap to the 41.0% peer average would convert roughly 14 one-time buyers into repeat customers.

Recommendation summary, ranked by value

#	Action	Owner	Expected impact
1	Cap discounts at 15% and move the spend into threshold-based bundles	Pricing / Trade Marketing	Rs 769,336 gross profit
2	Rebalance category budget from Electronics toward higher-margin lines	Category Management	Margin mix improvement
3	Delist or renegotiate the 16 sub-1% SKUs	Merchandising	Cost and complexity reduction
4	Retention programme in Salem before further acquisition spend	CRM / Regional Sales	Approx 14 buyers converted to repeat

Close with your own limitations

Stating what your analysis cannot conclude is a strong signal, not a weak one. For this dataset: returns are excluded so net margin is overstated; there is no marketing-spend data so no true ROI or CAC can be computed; the Rs 370,806 in revenue sitting under the "Unknown" region after cleaning limits regional precision; and discount effect is observational, not causal - proving it needs a controlled A/B test.

07 Package it so it is actually reviewable

A repository turns the work into a link you can put on a CV, and forces the discipline of naming and ordering your artifacts.

```
Sales_Performance_Analysis/
|-- README.md                                <- the case study, most read file
|-- Sales_Performance_Analysis_Project_Guide.pdf
|-- data/
|   |-- raw_sales_data.csv                  1,626 rows, defects intact
|   |-- cleaned_sales_data.csv             1,577 rows + calculated columns
|   |-- Sales_Data_Raw.xlsx                workbook for the Excel stage
|   +-- data_quality_log.csv              the cleaning audit trail
|-- sql/
|   |-- 01_schema_and_load.sql             DDL + load + integrity checks
|   |-- 02_business_questions.sql         the question ledger
|   +-- 03_advanced_analysis.sql          windows, Pareto, discounts, RFM
+-- scripts/
    |-- generate_sales_dataset.py          reproducible generator
    |-- minipdf.py                        dependency-free PDF writer
    +-- build_project_guide_pdf.py        builds this document
```

README order that works: Business problem, Data and assumptions, Approach, Key findings (dashboard screenshot at the top), Recommendations, Tools, Limitations.

Do not skip this

The highest-leverage artifact is a one-page executive summary. It is what a hiring manager will actually read, and almost no candidate produces one.

08 Write the LinkedIn post

Lead with the finding, not the tool. Tools get you filtered; findings get you remembered. Weak version: "Made a Power BI dashboard using SQL."

```
Sales Performance Analysis | Excel . SQL . Power BI

A retailer's revenue was flat for two quarters while order volume held steady. Leadership needed to know where to put the next promotional budget.

I analysed 1,577 transactions end to end. The problem was not demand. It was discounting.

What I found
-> Discounts above 20% produced a 26% LOWER average order value than undiscounted orders, while margin fell from 36.1% to 5.4%
-> 10 of 32 products drove 80% of revenue; 16 SKUs contributed under 1% each
-> Electronics generated 62% of revenue at only 14.9% margin - the biggest category was the weakest one
-> Salem had strong acquisition but a 12.5% repeat rate against a 41.0% average

What I recommended
-> Cap discounts at 15%, shift spend to bundle thresholds -> Rs 769,336 gross profit recovered at constant volume
-> Rebalance category budget toward higher-margin lines
-> Fix retention in Salem before spending more on acquisition there

How: Excel and Power Query for cleaning (every exclusion rule documented, 3.0% of rows), SQL with CTEs and window functions for the analysis, Power BI star schema with DAX measures for the dashboard.

The lesson: the interesting answer was not in the revenue chart.
```

It was in the margin column nobody asked for.

Full breakdown and code: <repo link>

#BusinessAnalyst #SQL #PowerBI #DataAnalytics #Excel

Post mechanics

- ? Dashboard screenshot as the image, or a three-slide carousel: problem, dashboard, recommendations
- ? Short lines and visible whitespace - no wall of text
- ? Put the repo link in the post body, and reply to every comment in the first two hours

09 Resume bullets

Concise version - one line, for a packed resume

Sales Performance Analysis | Excel, SQL, Power BI - Analysed 1,577 sales transactions to identify a Rs 769,336 margin leak from over-discounting; built a Power BI dashboard and delivered 4 costed recommendations targeting a 10% lift in average order value.

Expanded version - for a projects section

Sales Performance Analysis
Excel (Power Query) | SQL (CTEs, window functions) | Power BI (DAX, star schema)

- Cleaned and validated 1,626 raw sales records, documenting 7 data-quality rules affecting 3.0% of rows to make the analysis reproducible and auditable
- Built SQL analyses covering revenue trend, Pareto concentration, discount effectiveness and customer repeat behaviour across 32 SKUs and 639 customers
- Designed a single-page executive Power BI dashboard with month-on-month growth, margin and AOV measures across 3 filter dimensions
- Found that discounts above 20% reduced average order value by 26% while cutting margin from 36.1% to 5.4%; recommended a discount cap and bundle strategy projected to recover Rs 769,336 in gross profit

Every bullet carries a number and an outcome. That is the whole formula.

10 Defend it in the interview

These questions are close to guaranteed. Prepare the answers before you post the project.

Question they will ask	What they are really testing
"Walk me through the business problem you were solving."	Whether you lead with the problem or with the tool
"How did you decide which rows to remove, and how might that bias the result?"	Analytical honesty and data-quality judgement
"Your biggest region by revenue - is it your most valuable region?"	Whether you separate revenue from profit and lifetime value
"How confident are you in the Rs 769,336 estimate? What assumptions does it rest on?"	Whether you understand the limits of your own model
"Sales says your discount cap will cost them volume. Your response?"	Stakeholder pushback and commercial judgement
"Could the discount effect be caused by something else?"	Correlation versus causation; do you reach for an A/B test
"What would you analyse next and what data would you need?"	Strategic thinking beyond the immediate task
"Why a star schema instead of one flat table?"	Technical depth in the modelling layer

The question that separates candidates

Have a crisp answer ready for "What surprised you?" Genuine surprise is evidence that you did the analysis rather than reverse-engineering charts from a tutorial.

11 Mistakes that quietly kill these projects

Mistake	Fix
Recommending on margin when the data only has revenue	Add a Unit Cost column before you analyse anything
Six dashboard pages and forty visuals	One decision-focused page; detail behind it
Chart titles like "Revenue by Month"	Title every visual with the finding it shows
Insights that only restate a number	Every insight ends in an action with a rupee impact
Using the Kaggle dataset everyone else used	Generate your own, or pick an unusual angle
No record of cleaning decisions	Keep the Data Quality Log as a deliverable
Claiming precision the data cannot support	State assumptions and limitations explicitly
Leading the LinkedIn post with the tools	Lead with the finding; tools go at the bottom
Counting orders with COUNT(order_id) on a multi-line table	COUNT(DISTINCT order_id) - otherwise every KPI is inflated

12 Realistic timeline

Day	Work	Output
1	Write the project brief, generate the dataset with its built-in patterns	Brief + raw_sales_data.csv
2	Excel cleaning, Data Quality Log, calculated columns	Cleaned workbook
3	Load to SQL: write the question ledger	02_business_questions.sql
4	Advanced SQL: Pareto, discount bands, repeat rate	03_advanced_analysis.sql
5	Power BI model: relationships and DAX measures	sales_dashboard.pbix
6	Dashboard layout, formatting, insight annotations	Dashboard screenshot
7	README, one-page executive summary, LinkedIn post	Published repo and post

One focused week, or two weekends if you are working. The value is not in the hours - it is in finishing the arc from business problem all the way through to a costed recommendation.

A Appendix - verified dataset profile

Computed from `data/cleaned_sales_data.csv` when this PDF was built. Use these as the reference values when you rebuild the analysis yourself - if your SQL or Power BI numbers disagree with this table, something in your model is wrong.

Metric	Value
Rows (raw / cleaned)	1,626 / 1,577 (3.0% removed)
Distinct orders	1,145
Units sold	2,553
Distinct customers	639
Distinct products	32
Net revenue	Rs 19,333,556
Gross profit	Rs 4,450,106
Overall gross margin	23.0%
Average order value	Rs 16,885
Overall repeat rate	35.4%
Date range	2025-09-01 to 2026-08-29
Peak month	2025-10 (Rs 2,526,792)
Weakest month	2026-01 (Rs 968,900)

Monthly net revenue and month-on-month growth

Month	Net revenue	MoM	Month	Net revenue	MoM
2025-09	Rs 1,240,172		2026-03	Rs 1,264,332	+23.0%
2025-10	Rs 2,526,792	+103.7%	2026-04	Rs 1,690,458	+33.7%
2025-11	Rs 2,286,674	-9.5%	2026-05	Rs 1,364,884	-19.3%
2025-12	Rs 1,981,300	-13.4%	2026-06	Rs 1,599,144	+17.2%
2026-01	Rs 968,900	-51.1%	2026-07	Rs 1,728,395	+8.1%
2026-02	Rs 1,028,106	+6.1%	2026-08	Rs 1,654,399	-4.3%

Full product ranking by net revenue (cumulative share shows the Pareto curve)

#	Product	Net revenue	Cum.	#	Product	Net revenue	Cum.
1	Laptop	Rs 4,199,284	21.7%	17	Cookware Set	Rs 167,476	93.4%
2	Phone	Rs 2,696,564	35.7%	18	Jacket	Rs 163,717	94.2%
3	Smart TV	Rs 2,536,568	48.8%	19	Formal Shirt	Rs 135,982	94.9%
4	Sofa Set	Rs 1,523,798	56.7%	20	Power Bank	Rs 129,856	95.6%
5	Tablet	Rs 1,107,197	62.4%	21	Saree	Rs 115,212	96.2%
6	Bed Frame	Rs 866,620	66.9%	22	Jeans	Rs 110,154	96.8%
7	Wardrobe	Rs 782,772	70.9%	23	Dinner Set	Rs 100,827	97.3%
8	Smartwatch	Rs 724,279	74.7%	24	Backpack	Rs 90,979	97.7%
9	Water Purifier	Rs 702,700	78.3%	25	Kurta	Rs 83,705	98.2%
10	Microwave	Rs 515,492	81.0%	26	T-Shirt	Rs 76,479	98.6%
11	Office Chair	Rs 463,312	83.4%	27	Pressure Cooker	Rs 72,538	98.9%
12	Study Table	Rs 451,376	85.7%	28	Sunglasses	Rs 60,072	99.3%
13	Mixer Grinder	Rs 419,627	87.9%	29	Wallet	Rs 48,028	99.5%
14	Bluetooth Speaker	Rs 316,510	89.5%	30	Phone Case	Rs 34,838	99.7%
15	Headphones	Rs 312,929	91.1%	31	Laptop Sleeve	Rs 32,218	99.9%
16	Bookshelf	Rs 264,500	92.5%	32	Belt	Rs 27,948	100.0%